

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

December 2016

EC203 Solid State Electronics

Date: 20.12.2016, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)**
1. A semiconductor has temperature coefficient of resistance. [01]
(i) positive (ii) zero (iii) negative (iv) none of the above
 2. The strength of a semiconductor crystal comes from [01]
(i) forces between nuclei (ii) forces between protons (iii) electron-pair bonds (iv) none of the above
 3. A pentavalent impurity has valence electrons. [01]
(i) 3 (ii) 5 (iii) 4 (iv) 6
 4. An n-type semiconductor is [01]
(i) positively charged (ii) negatively (iii) electrically neutral (iv) none of the above
 5. A hole and electron in close proximity would tend to [01]
(i) repel each other (ii) attract each other (iii) have no effect on each other (iv) none of the above
 6. The random motion of holes and free electrons due to thermal agitation is [01]
called
(i) diffusion (ii) pressure (iii) ionisation (iv) none of the above
 7. In the depletion region of a pn junction, there is a shortage of [01]
(i) acceptor ions (ii) holes and electrons (iii) donor ions (iv) none of the above

Q - 2 (a) Draw and explain circuit diagram of bridge type full wave rectifier. Derive its mathematical equation for maximum efficiency. [07]

(b) What is stabilization in transistor? Describe the need of stabilization in detail. [07]

OR

Q - 2 (a) An ac supply of 230v is applied to half wave rectifier through transformer of turn ratio 10:1. find dc output voltage, PIV, rectification efficiency. Assume diode to be ideal. [07]

- (b) How transistor load line analysis bifurcates three different regions on operation of transistor. [07]
- Q - 3 (a) Define and Derive expression of R_i , R_o , A_i , A_v , and A_p for CC amplifier. [07]
- (b) State and explain Miller's theorem. [07]

OR

- Q - 3 (a) Explain h parameter model of CE amplifier and state the typical value of h parameter for the same. [07]
- (b) Explain the operation of emitter follower amplifier. Why is it called emitter follower? [07]

SECTION – II

- Q - 4 (a) Write a note on hybrid – pi capacitance. [07]
- Q - 5 (a) How much valid hybrid pie model and Explain variation in different parameter effective table. [07]
- (b) Describe construction and working of junction field effect transistor with necessary diagram. [07]

OR

- Q - 5 (a) Explain hybrid pie conductance and derive its transconductance equation. [07]
- (b) Explain JFET parameter with its detail. Derive relationship between them. [07]
- Q - 6 (a) Explain the following terms: [07]
- (i) Frequency response (ii) Decibel gain (iii) Bandwidth gain
- (b) Compare all types of multistage amplifier with its technical parameter. [07]

OR

- Q - 6 (a) Draw and explain circuit diagram of RC coupled multi stage amplifier. Enlist its advantages and disadvantages. [07]
- (b) What do you understand by multi stage transistor amplifier? Mention its need. [07]
